CHAPTER 4

PRESENTATION OF DATA

CHAPTER ESSENCE AT A GLANCE:
Master structural data communication: textual narrative presentation, formal statistical tables and parts of a table, diagrammatic bar or pie charts, and graphic histograms, frequency polygons, and ogives.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Three Presentation Formats:** Textual presentation (embedding numbers into narrative prose), Tabular presentation (systematic rows and columns), and Diagrammatic and Graphic presentation (visual geometric forms).

• **Essential Parts of a Statistical Table:** Table Number, Title, Captions (column headings), Stubs (row headings), Body of the table (numerical cells), Unit of measurement, Source note, and Footnotes.

• **Diagrammatic Geometric Visuals:** Bar diagrams (Simple, Subdivided, Percentage, Multiple bars) and Pie diagrams (circle divided into sectors proportional to 360 degrees).

• **Graphic Visualizations of Distributions:** Histograms (adjacent rectangles over continuous intervals), Frequency Polygons, Frequency Curves, and Cumulative Frequency Curves (Ogives used to locate median).

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Pie Chart Component Angle } (\theta_i) = \frac{\text{Component Value}}{\text{Total Aggregate Value}} \times 360 degrees$$

Circular Sectoral Angular Proportion Formula

$$\text{Percentage Bar Component} = \frac{\text{Individual Category Value}}{\text{Total Group Aggregate}} \times 100$$

Subdivided Percentage Bar Normalization

$$\text{Ogive Intersection Point} \iff \text{Less-than Ogive} \cap \text{More-than Ogive} \rightarrow \text{Median } (M)$$

Graphic Median Derivation Invariant

$$\text{Histogram Modal Peak} \iff \text{Highest Rectangle Apex} \rightarrow \text{Mode } (Z)$$

Graphic Mode Derivation Invariant

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Histograms can be drawn only for continuous grouped frequency distributions; they can NEVER be drawn for discrete or individual series.

• **Alert:** The intersection point of 'Less-than' and 'More-than' ogives projected on the X-axis gives the exact value of the Median.

• **Alert:** Pie diagram components must always sum to exactly 360 degrees (or 100 percent in percentage terms); always convert component values to angles first.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is Tabular Presentation of data? State two primary advantages of tables. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The systematic arrangement of statistical data in horizontal rows and vertical columns. [1 Mark]

• **Advantages:** (1) Facilitates rapid visual comparison between different variables, (2) Compacts large volumes of data into minimal space. [1 Mark]

2. Distinguish between a Caption and a Stub in a statistical table. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Caption:** The heading given to a vertical column in a table, describing the column contents. [1 Mark]

• **Stub:** The heading given to a horizontal row in a table, describing the row categories. [1 Mark]

3. Enumerate any six essential parts of a formal Statistical Table. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Core Elements 1, 2, 3:** (1) Table Number, (2) Title of the Table, (3) Captions (column headings). [1.5 Marks]

• **Core Elements 4, 5, 6:** (4) Stubs (row headings), (5) Body of the table (numerical figures), (6) Source Note and Footnotes. [1.5 Marks]

4. Convert the following expenditures of a family into degrees for constructing a Pie Diagram: Food Rs 6,000; Clothing Rs 3,000; Rent Rs 4,000; Education Rs 3,000; Savings Rs 2,000. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Total Expenditure:** Rs 6,000 + Rs 3,000 + Rs 4,000 + Rs 3,000 + Rs 2,000 = Rs 18,000 = 360 degrees. (1 rupee = 360 divided by 18,000 = 0.02 degrees). [1 Mark]

• **Degree Calculations:** Food = (6,000 divided by 18,000) * 360 = 120 degrees; Rent = (4,000 divided by 18,000) * 360 = 80 degrees. [1 Mark]

• **Remaining Degrees:** Clothing = 60 degrees; Education = 60 degrees; Savings = 40 degrees. Total = 120 + 80 + 60 + 60 + 40 = 360 degrees. [1 Mark]

5. Distinguish between a Bar Diagram and a Histogram across three parameters. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Spacing:** Bar diagrams have uniform gaps between adjacent bars; Histograms have contiguous rectangles with zero gap between class intervals. [1 Mark]

• **Data Type:** Bar diagrams represent discrete or categorical data; Histograms represent continuous frequency distributions only. [1 Mark]

• **Dimensional Character:** Bar diagrams are one-dimensional (only height matters); Histograms are two-dimensional (area of rectangle is proportional to frequency). [1 Mark]

6. Explain how the Median and the Mode can be determined graphically. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Graphic Median:** Plot the 'Less-than' and 'More-than' ogives on the same graph; the X-coordinate of their point of intersection gives the Median value. [1.5 Marks]

• **Graphic Mode:** Identify the highest rectangle in a histogram; draw diagonal lines from its top corners to adjacent rectangles; the X-coordinate of the intersection point gives the Mode. [1.5 Marks]

7. Explain the four major types of Bar Diagrams: Simple bar diagram, Multiple bar diagram, Subdivided (component) bar diagram, and Percentage bar diagram. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Simple Bar Diagram:** Displays one single variable over time or across categories using equidistant vertical or horizontal bars. **[1 Mark]**

• **Multiple Bar Diagram:** Displays two or more interrelated variables side-by-side (e.g. imports and exports over successive years). **[1 Mark]**

• **Subdivided (Component) Bar Diagram:** Total magnitude of a bar is partitioned into constituent parts (e.g. total cost divided into material, labor, overheads). **[1 Mark]**

• **Percentage Bar Diagram:** Components are converted into percentages where each bar is drawn to a uniform 100 percent height, facilitating relative comparisons. **[1 Mark]**